

Joint Segmentation and Differential Diagnosis of Thyroid Nodule in Contrast-Enhanced Ultrasound Images

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Abstract—Objective: Microvascular perfusion can be observed in real time with contrast-enhanced ultrasound (CEUS), which is a novel ultrasound technology for visualizing the dynamic patterns of parenchymal perfusion. Automatic lesion segmentation and differential diagnosis of malignant and benign based on CEUS are crucial but challenging tasks for computer-aided diagnosis of thyroid nodule. **Methods:** To tackle these two formidable challenges concurrently, we provide Trans-CEUS, a spatial-temporal transformer-based CEUS analysis model to finish the joint learning of these two challenging tasks. Specifically, the dynamic swin-transformer encoder and multi-level feature collaborative learning are combined into U-net for achieving accurate segmentation of lesions with ambiguous boundary from CEUS. In addition, variant transformer-based global spatial-temporal fusion is proposed to obtain long-distance enhancement perfusion of dynamic CEUS for promoting differential diagnosis. **Results:** Empirical results of clinical data showed that our Trans-CEUS model achieved not only a good lesion segmentation result with a high Dice similarity coefficient of 82.41%, but also superior diagnostic accuracy of 86.59%. **Conclusion & significance:** This research is novel since it is the first to incorporate the transformer into CEUS analysis, and it shows promising results on dynamic CEUS datasets for both segmentation and diagnosis tasks of the thyroid nodule.

Index Terms—Contrast-enhanced ultrasound, thyroid nodule, long-distance information, joint learning.

I. INTRODUCTION

THYROID nodule disease is an increasingly prevalent clinical disease, which has a dramatic increase in morbidity and mortality over the past few years. While a clinical diagnosis of thyroid nodules is common, only between 7%-15% of those patients may eventually develop thyroid cancer [1]. Consequently, distinguishing between thyroid cancer and thyroid nodules is a crucial clinical issue. Differentiating benign from malignant thyroid nodules is currently accomplished mostly using fine needle aspiration cytology (FNAC) and ultrasound imaging [2]. According to the available literature, the FNAC examination, which involves many punctures, can cause significant damage to the thyroid gland and has a considerable turnaround time between the sampling and the diagnostic result [3]. Consequently, noninvasive ultrasound imaging is the popular choice for differential diagnosis of thyroid nodules.

Different from B-mode ultrasound (US) or grey scale US, dynamic contrast enhanced ultrasound (CEUS) as a kind of functional imaging can implement the real-time and continuous observation of the whole process of circulating intra-tumor particles. Due to its non-radiation, minimal cost, and convenience of use, dynamic CEUS has achieved widespread application in clinical disease diagnosis and interventional treatment [4]. In practice, by injecting contrast agents intravenously, CEUS can effectively show the dynamic enhancement effect of arterial phase in real time [5], [6] and display the blood flow at the level of microcirculation perfusion of tissues and organs, which offers the dynamic observation of microvascular perfusion. Consequently, dynamic CEUS imaging contains the microvascular perfusion information related to lesion, which undoubtedly can help precise lesion diagnosis [7].

The thyroid is a highly microvascular organ, and both benign and malignant thyroid nodules have distinct vascular architectures. Differentiating benign from malignant thyroid nodules relies on their circulatory architecture, which are themselves unique. In order to achieve a differential diagnosis between benign and malignant lesions, physicians must carefully study the dynamic CEUS video and concentrate on the lesion's location throughout the diagnostic procedure. However, the clinical experience of the radiologists has a significant impact on the

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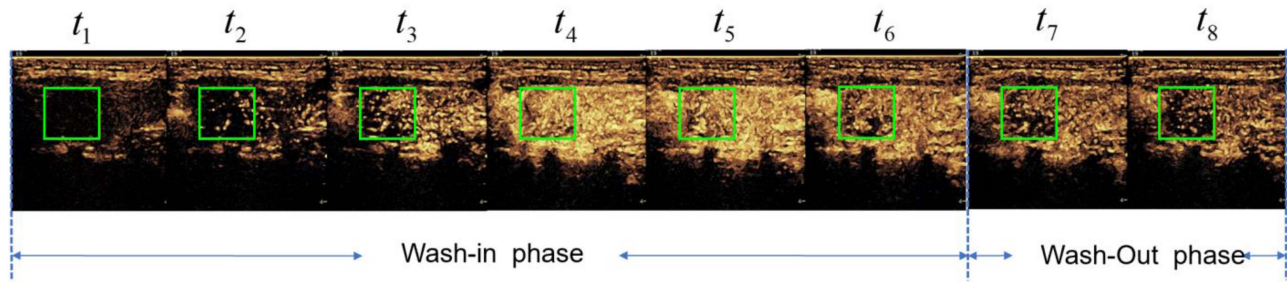


Fig. 1. This CEUS video sequence records the dynamic changes of microcirculation of thyroid nodules after injecting ultrasound contrast agents. We can have the following observations: (1) the lesion area (green box) and the background area are difficult to differentiate; (2) the image frames of CEUS sequence contain the wash-in and wash-out phases, thus, obtaining long-distance related information from CEUS sequence is difficult.

accuracy of the diagnosis, and they may easily make a mistaken or erroneous determination. In addition, manual lesion segmentation from CEUS is a laborious and time-consuming operation that is required for clinical diagnosis.

A growing number of studies in recent years have shown promise for the application of machine learning techniques in the analysis of CEUS video to aid disease detection [8] for a wide range of tissues. Specially, some studies proposed to use the machine learning methods to analyze CEUS images [9], [10] or use ultrasound image for breast cancer diagnosis [11], [12]. However, several obstacles exist for automated segmentation and illness identification with machine learning due to the unique nature of CEUS imaging modalities, including the following. (1) There is no obvious difference between the lesion area and the normal area in the CEUS, and the boundary of the lesion is ambiguous, which increases the difficulty of automatic segmentation and benign and by using CEUS (as shown in Fig. 1).

(2) Radiologists need to analyze the entire dynamic enhancement patterns of CEUS perfusion for revealing the blood supply of thyroid from wash-in to wash-out (as shown in Fig. 1). Radiologists evaluate long-distance enhancement information and dynamic perfusion patterns to determine if an image is benign or malignant; consequently, long-distance enhancement information must be analyzed and evaluated for the diagnosis model to replicate their diagnostic processes [13].

The purpose of this paper is to propose a novel CEUS analysis model that can perform automatic segmentation and diagnosis in parallel. The innovations behind our model include: 1) we propose the swin-transformer [14] based encoder with the collaborative learning of the low-level and high-level features (CLLH) to finish the segmentation of lesions with ambiguous boundary from CEUS. 2) the mechanisms of spatial temporal separation aggregation (STSA) decoder and global spatial-temporal fusion (GSTF) are developed to obtain long-distance enhancement information for analyzing dynamic CEUS data. In addition, dynamic CEUS analysis has shown success with the concept of jointing learning for the segmentation and differential diagnosis tasks.

II. RELATED WORKS

In this section, we focus on introducing some related studies about the computer-aided diagnosis by using the CEUS sequence. On the one hand, numerous studies on the topic

of lesion diagnosis using CEUS video analysis are presented. On the other hand, considering that transformer mechanism is used for CEUS analysis in this study, therefore, the related works about transformers based medical image analysis are also introduced.

A. CEUS Analysis Method

Traditional CEUS analysis methods obtain some functional parameters (i.e., nodule arrival time, time to peak, peak intensity, etc.) through the time intensity curve (TIC) of the lesion area to finish the computer-aided diagnosis [15]. The morphological information of a lesion, which is essential for diagnosing benign and malignant lesions, is typically disregarded by these techniques. Currently, most of the researches about CEUS-based automatic lesion detection and diagnosis mainly use the deep CNN to extract the spatial-temporal features of CEUS sequences. By using the Resnet50 and VGG16 classification network to extract features from CEUS images, Qin et al. realized the transfer learning-based grading diagnosis of colorectal cancer [16]. Yang et al. emphasized the importance of the time sequence of CEUS, and combined the CNN with a Temporal Sequence Regression Mechanism (TSRM) to extract the dynamic enhanced feature of the lesion area for automatic diagnosis [17]. Based on the inherent diagnostic methods of clinicians, Wan et al. [18] proposed a hierarchical temporal attention CNN network which used the spatial feature enhancement and fused the local and global temporal feature for disease diagnosis based on dynamic CEUS images. Furthermore, by using the CEUS domain knowledge and combing the US modality, Chen et al. [19] proposed domain-knowledge-guided temporal attention module to achieve the breast cancer diagnosis. However, obtaining the lesion area necessary for disease diagnosis still requires laborious manual labelling using the methods. While the transformer has outperformed CNNs in a number of visual representation tasks, no previous studies have applied the transformer to the problem of autonomous processing of CEUS data.

In the field of lesion segmentation, few studies have conducted in-depth research on CEUS segmentation. As a representative study, Wan et al. [20] proposed a CEUS-Net which could achieve thyroid and breast lesion segmentation using the multi-channel convolution and feature-reweighted dense module. However, this model cannot obtain the global temporal dimension information using the channel attention for temporal fusion. We

also think that the output of a good segmentation model can be of great assistance in the benign and malignant diagnosis task, although there has been little study of combining these two processes for CEUS analysis [21].

B. Temporal and Spatial Transformer

Currently, in the realm of computer vision, transformer has shown robust visual representation skills, particularly when it comes to modeling and evaluating the temporal sequence [22]. In addition, the transformer has shown promoting results than CNN in some visual representation tasks, but no researchers have applied the transformer to the task of CEUS automatic analysis with small-scale dataset. Despite the common belief that a large number of datasets are necessary for training a transformer to produce satisfactory results, transformer is increasingly being used for medical image analysis tasks with limited data. For example, improved transformer models such as Trans-unet [23] and swin-unet [24] have been proposed for segmenting 2D and 3D medical images, including Computed Tomography (CT), Magnetic Resonance (MR), etc. Compared with CT and MR medical imaging modalities, CEUS videos are explicit temporal sequences, which contain dependencies information among image frames [25]. Therefore, transferring the developed transformer models of CT, MR images to CEUS video analysis tasks directly is difficult to achieve excellent results. In addition, some studies propose to use transformer technique for ultrasound image analysis [26], [27], [28]. For example, Qu et al. proposed the vision transformer module to learn the global relationship among different regions and enhance the relevant local features for breast cancer diagnosis [26]. Plotka et al. proposed the residual transformer module for direct estimations of fetal weights from 2D+t spatio-temporal ultrasound video scans [28]. To our knowledge, specific transformer model proposed for the CEUS analysis task has not been exploited.

In this study, we propose a novel spatial-temporal transformer-based CEUS analysis model called Trans-CEUS for finishing the automatic lesion segmentation and the differential diagnosis simultaneously. In contrast to the existing transformer architecture, a number of enhancements are presented to overcome the challenges encountered during the CEUS analysis process. For accurate lesion detection from CEUS, a swin-transformer encoder mechanism with collaborative learning of low-level and high-level characteristics is first designed. Second, for extracting the long-distance enhanced perfusion from the dynamic CEUS picture, our CEUS analysis model Trans-CEUS proposes spatial temporal separation aggregation and global spatial-temporal fusion methods. The experimental findings demonstrate that Trans-CEUS performs well on CEUS datasets for segmentation and identification of thyroid nodules.

III. METHOD

A. Overview

Our proposed Trans-CEUS model mainly consists of two platforms, as shown in Fig. 2. Given a dynamic CEUS sequence, a set of CEUS image frames $I = \{I_1, I_2, I_3 \dots, I_T\}$ that cover

the wash in and wash out phases are used as the input of the Trans-CEUS. Trans-CEUS model uses a spatial backbone Resnet18 as a 2D frame-level spatial feature extractor for the purpose of extracting the fundamental low-level characteristics. And the extracted frame-level spatial feature is inputted into the lesion segmentation module (Platform1) and the differential diagnosis module (Platform 2).

In Platform 1, the dynamic swin-transformer encoder is combined with the spatial temporal separation aggregation (STSA) decoder into Unet [29] structure for achieving accurate lesion identification from CEUS. In addition, the collaborative learning module between the low-level and high-level features (CLLH) is developed to exploit the strong complementarity among various information for promoting the identification of small-size lesions with ambiguous boundary.

Thyroid nodule benign/malignant discrimination in Platform 2 is achieved by including a segmented lesion mask into a spatial feature. A spatial feature extractor is employed for both the lesion segmentation and differential diagnostic tasks, with further fine-tuning performed using a CNN based local fine-tuning (LFT) unit. In addition, the global spatial-temporal fusion (GSTF) based on variant transformer is proposed to integrate the dynamic temporal features for benign and malignant diagnosis.

B. Swin-Transformer Based Lesion Segmentation

Due to their unexpected ability to extract spatial-temporal characteristics from natural images, transformers have become more popular in numerous computer vision tasks. However, the straightforward application of the transformer model to CEUS video processing tasks often confronts a number of obstacles. Firstly, for the CEUS video analysis task, we need to dynamically update the receptive field to adapt to the dynamic feature representation in different periods (perfusion period, scrubbing period), and make full use of the long-distance related information of CEUS. Secondly, over-fitting is readily triggered by a small-scale CEUS dataset. In this paper, the swin-transformer based lesion segmentation platform is developed by using the Spatial Temporal Separation Aggregation (STSA) decoder unit and the collaborative learning module of the low-level and high-level features (CLLH). Specifically, there is a dynamic swin-transformer encoder and a STSA decoder in the lesion segmentation platform. The feature maps from encoder and decoder parts are connected into a U-net architecture [29] through skip connections. Here, the dynamic swin-transformer encoder uses the dynamic sliding window mechanism for better adapting to the CEUS video analysis task and reducing the computational cost. Additionally, the CLLH module is proposed to reduce the noise impact of low-level features and aid high-level features in more accurately identifying the tiny lesion region.

The input $2D+t$ CEUS image sequence is denoted as $I = \{I_t\}_{t=1 \dots T} = \{I_1, I_2, I_3 \dots I_T\} \in \mathbb{R}^{C \times T \times H \times W}$, where $C = 3$ represents the number of channels in each frame, T represents the CEUS sequence length, H and W represent the length and width of each frame, respectively. Firstly, the Resnet18 without downsampling is used as the frame-level feature extractor to extract the feature map $f_t^{CEUS} \in \mathbb{R}^{C \times H \times W}$ for each CEUS frame

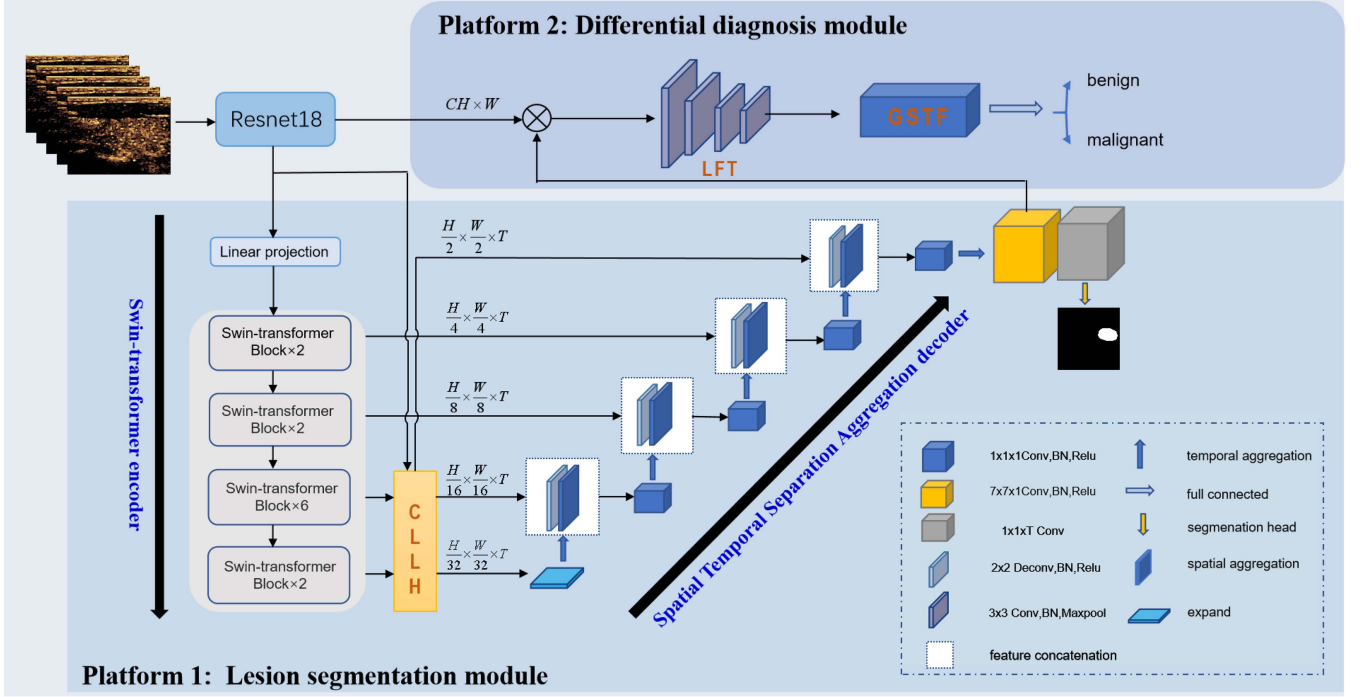


Fig. 2. Framework of the proposed spatial-temporal transformer-based CEUS analysis model (Trans-CEUS). The framework consists of two platforms including the lesion segmentation module and the diagnosis module. These two modules share the spatial feature extractor of Resnet18. In Platform 1, the dynamic swin-transformer encoder with the collaborative learning of low-level and high-level features (CLLH) is combined with the spatial temporal separation aggregation (STSA) decoder to achieve accurate lesion segmentation. In Platform 2, inspired by the CNN-transformer hybrid mechanism, the CNN based local fine-tuning (LFT) and variant transformer-based global spatial-temporal fusion (GSTF) are used together to finish precise diagnosis.

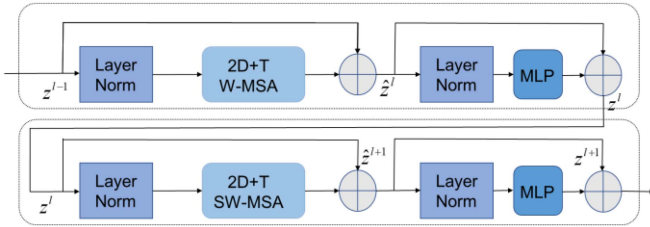


Fig. 3. Details of the swin-transformer block.

I_t . Thus, the feature map of the whole CEUS sequence is denoted as $F^{CEUS} = [f_1^{CEUS}, f_2^{CEUS}, \dots, f_T^{CEUS}] \in \mathbb{R}^{C \times T \times H \times W}$.

1) Swin-Transformer Encoder: With the feature map $F^{CEUS} \in \mathbb{R}^{C \times T \times H \times W}$ of the whole CEUS sequence as the input of the linear projection part, we treat each patch of size $C \times T' \times H' \times W'$ as a token, where $T' = 1, H' = 4, W' = 4$. The linear projection part can acquire a sequence of tokens with size of $\frac{T}{T'} \times \frac{H}{H'} \times \frac{W}{W'}$. Inspired by the video swin-transformer [14], we use the shifted windows (SW) strategy to explore the connections between tokens for effective feature encoder during the swin-transformer block.

Swin-transformer block: At layer l , the windows size is $P \times M \times M$, and a token is divided into $\lceil \frac{T'}{P} \rceil \times \lceil \frac{H'}{M} \rceil \times \lceil \frac{W'}{M} \rceil$ non-overlapping windows. At the $l+1$ layer, the partitioned windows are shifted by $\frac{P}{2} \times \frac{M}{2} \times \frac{M}{2}$ voxels. As shown in Fig. 3, the

output of continuous l and $l+1$ layers are calculated as follows:

$$\hat{z}^l = W-MSA(LN(z^{l-1})) + z^{l-1}$$

$$z^l = MLP(LN(\hat{z}^l)) + \hat{z}^l$$

$$\hat{z}^{l+1} = SW-MSA(LN(z^l)) + z^l$$

$$z^{l+1} = MLP(LN(\hat{z}^{l+1})) + \hat{z}^{l+1}$$

where $W-MSA$ and $SW-MSA$ denotes the windows based multi-head self-attention (MSA) using regular and shifted window partitioning configurations, respectively; $\hat{z}^l$ and $\hat{z}^{l+1}$ denote the outputs of $W-MSA$ and $SW-MSA$. LN and MLP are the linear normalization and multi-layer perception operations. Here, we follow the study in [8] and include a relative position bias $B \in \mathbb{R}^{P^2 \times M^2 \times M^2}$ into self-attention computing, as follows:

$$Attention(Q, K, V) = SoftMax(QK^T / \sqrt{d} + B)V$$

where Q, K, V are the queries, keys, and values, respectively; d is the dimension of query and key. As depicted in Fig. 2, our encoder path uses 4 stages, which contains 22,62 transformer blocks respectively to acquire the corresponding encoder output $\{F_1, F_2, F_3, F_4\}$.

2) Spatial Temporal Separation Aggregation (STSA) Decoder: Before the STSA decoder, the encoder output $\{F_1, F_2, F_3, F_4\}$ is input into the collaborative learning module of low-level and high-level features (CLLH) to enhance

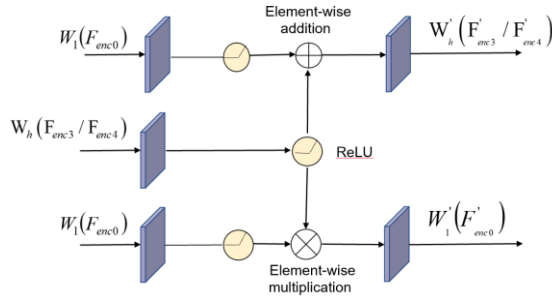


Fig. 4. Details of the collaborative learning module of low-level and high-level features.

the feature representation. Inspired by R(2+1)D model, STSA decoder factorizes the traditional 3D convolutional filters into separate spatial and temporal components. Specially, the encoded features F_i and F_{i+1} are concatenated and performed a 2×2 upsampling operation in the spatial dimension. In addition, in the temporal dimension, a convolution operation with kernel size of $1 \times 1 \times 3$ is used to finish the temporal aggregation. After the STSA decoder, the corresponding mask sequence I_{mask} is acquired, and we use a spatial convolution with size of $7 \times 7 \times 1$ and a temporal convolution with size of $1 \times 1 \times T$ to finish mutual complementary of the mask sequence and accurately locate the lesion.

a) Collaborative learning module of low-level and high-level features: Since there is an inequity between the background and lesion regions, tiny lesions are readily eliminated during the continual downsampling process. To avoid this limitation, low-level features W_l of CEUS should receive more attention because low-level features W_l has higher space resolution. However, low-level features W_l of CEUS cannot be directly applied to segmentation tasks due to it may contain noise. On the contrary, high-level features W_h have filtered out a lot of noise by continuous down-sample, but the small lesion area is easily eliminated during the continuous down-sampling.

In medical image analysis, small lesion segmentation commonly relies on shallow features [30], because they have higher resolutions and contain richer details compared with deep ones [31]. Unfortunately, low-level features are too noisy to be used directly [30]. Therefore, we design the collaborative learning module of low-level and high-level features (CLLH), as shown in Fig. 4, to eliminate the noise effect of low-level features and help high-level features to better identify the small lesion area [32]. For example, we regard the third stage output F_{enc3} of swin-transformer as high-level W_h and output F_{enc0} of Resnet18 as the low-level feature W_l , then the collaborative learning of W_h and W_l is shown in following equations:

$$W'_h = \sigma [\sigma(Down(W_l)) \oplus \sigma(W_h)]$$

$$W'_l = \sigma [\sigma(Up(W_h)) \otimes \sigma(W_l)]$$

where $Up(\cdot)$ and $Down(\cdot)$ represent upsampling and downsampling operations, respectively. σ represents reshaping feature maps to the same size. $\oplus$ and $\otimes$ represent element-wise addition and multiplication operations, respectively. We also regard the

fourth stage output F_{enc4} of swin-transformer as high-level W_h to complete collaborative learning and output F'_{enc4} . After CLLH, we can make better use of high-level features and obtain more useful low-level features.

C. Differential Diagnosis Task

For finishing the precise differential diagnosis by using dynamic CEUS, on the one hand, we propose to finish the joint segmentation and diagnosis task simultaneously, which uses the automatically segmented lesion region to promote the diagnosis task. On the other hand, malignancy and benignity predictions are made using a combination of local fine-tuning (LFT) for CNN and global spatial-temporal fusion (GSTF) for variant transformers.

The idea of combining segmentation and diagnosis is inspired by the imageomics-based differential diagnosis [33], [34]. In the approach of imageomics-based diagnosis, the lesion area is manually segmented, and then image characteristics are extracted from the segmentation region to forecast malignant and benign [35], [36]. Therefore, the input of the differential diagnosis task is the result of multiplying the original CEUS image feature map F^{CEUS} by the segmentation mask sequence I_{mask} . Crucially, two-task joint training allow us to learn a task-specific lesion method which is tuned to optimize the performance of differential diagnosis.

As shown in Fig. 2, the differential diagnosis task mainly comprises a CNN-based LFT unit and a variant transformer-based GSTF unit. Here, we directly use the spatial features obtained by resnet18 in the segmentation task as the benchmark. In addition, a four-layer channel-separable convolution network [37] is used in this LFT unit to fine-tune the extracted spatial features. The first and third layers contain the convolutions with 1×1 and 3×3 kernels, followed by BatchNorm and RELU activity function. About the second and fourth layers, we use the channel-separable convolutions with 3×3 kernels, followed by an average pooling layer with a stride of 2. This LFT unit not only achieves the effective spatial feature extraction, but also the CEUS feature map $F^{CEUS} \in \mathbb{R}^{C \times T \times H \times W}$ is reshaped into $F^* \in \mathbb{R}^{C \times T \times \frac{H}{16} \times \frac{W}{16}}$. The reshaped feature map $F^* \in \mathbb{R}^{C \times T \times \frac{H}{16} \times \frac{W}{16}}$ is embedded into $F^{**} \in \mathbb{R}^{T \times D}$, where $D = C \times \frac{H}{16} \times \frac{W}{16}$ by using the linear embedding layer and input into the GSTF unit.

1) Global Spatial Temporal Fusion (GSTF): Medical professionals frequently employ the whole CEUS sequence to detect the pattern of blood flow changes in tumors microvessels, allowing them to distinguish between benign and malignant tumors. Inspired by this clinical phenomenon, the global spatial temporal fusion (GSTF) unit based on variant transformer is proposed to extract the long-distance enhancement perfusion from the dynamic CEUS image effectively. Evidence shows that when applied to a small dataset, the transformer leads to excessive fitting. For categorization purposes, the partial structural deletion or alteration of the transformer was confirmed by Chen et al. [38]. In light of this, we develop the GSTF using a distinct kind of transformer. The attention operation and delete MLP unit are the only two operations allowed by the variant transformer

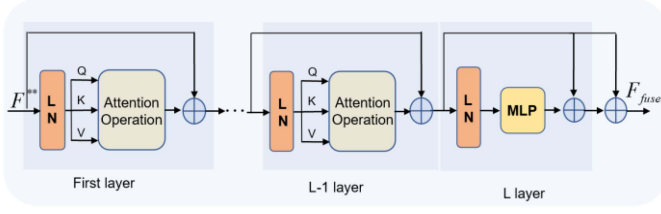


Fig. 5. GSTF with variant transformer which contains the linear normalization (LN), attention operation, and multi-layer perception (MLP).

blocks in GSTF's L-1 layer. As shown in Fig. 5, the final output feature F_{fuse} is expressed as follows:

$$\begin{aligned} cls_1 &= Att[LN(F^{**})] + F^{**} \\ &\dots \\ cls_{l-1} &= Att[LN(cls_{l-2})] + cls_{l-2} \\ cls_l &= MLP[LN(cls_{l-1})] + cls_{l-1} \\ F_{fuse} &= cls_{l-1} + cls_l \end{aligned}$$

where $Att[\cdot]$, $MLP[\cdot]$ and $LN(\cdot)$ denote the attention operation, multi-layer perception and linear normalization, respectively. $cls_{i, i \in \{1, 2, \dots, l\}}$ denotes the output of the i layer. Here, we use four attention layers in the GSTF unit of diagnosis task. Finally, the fused feature F_{fuse} is input into a fully connected layer to get a vector of $F_{cls} \in \mathbb{R}^{1 \times 2}$ for benign and malignant diagnosis.

IV. EXPERIMENTS

In this section, we demonstrate the effectiveness of our Trans-CEUS through a series of convincing experiments. Primarily, we briefly introduce the CEUS dataset of the thyroid nodule and the implementation details of the experiments. Next, we evaluate the lesion segmentation results of thyroid nodule from CEUS by using the Trans-CEUS model. Then, we present the performance of the diagnosis task. We conclude by assessing the benefits of combining the segmentation and diagnosis tasks for thyroid nodule.

A. Dataset and Implementation Details

1) Dataset: Our thyroid nodule dataset contained 163 consecutive patients who underwent thyroid nodule examination at Nanjing Drum Tower Hospital. All patients performed dynamic CEUS imaging examination by an experienced sonographer using an iU22 scanner (Philips Healthcare, Bothell, WA) equipped with a linear transducer L9-3 probe. These 163 cases in our dataset included 87 malignant nodules and 76 benign nodules. All patient data were collected legally, with their informed permission and IRB approval.

The ground truths of segmentation and diagnosis for thyroid nodule: On the one hand, percutaneous biopsies with subsequent pathological analysis were employed to establish the definitive cancer diagnosis or benign illness. On the other hand, to obtain the lesion masks for training the segmentation task,

radiologists with more than 10 years of experience manually annotated the pixel-level ground-truth of thyroid lesions.

2) Implementation Details: Our model was developed in Pytorch, and the corresponding code was executed using a single NVIDIA RTX 12 GB GPU. During the development of the Trans-CEUS model, a learning rate of 0.0001 and a total of 100 epochs were used. The Adam optimizer was utilized to fine-tune the network parameters continuously during the training process. The input CEUS video comprised complete wash-in and wash-out phases, and each frame had a resolution of 224 pixels $\times$ 224 pixels. To evaluate the model's performance, we implemented 5-fold cross-validation during the experiments. We used universal segmentation metrics, such as Dice and IOU indexes, to evaluate the segmentation performance. Additionally, we evaluated the results of the diagnosis task using four indexes, including the average accuracy (ACC), sensitivity (Se), specificity (Sp), false positive rate (FPR), and F1-score (F1).

We followed a two-stage approach to train our joint segmentation and diagnosis model. In the first stage, we used Platform 1 to achieve CEUS lesion segmentation, which helped our diagnosis model to accurately locate the lesion. In the second stage, we used the trained optimal model of Platform 1 to locate the lesion, and used the lesion area from the original CEUS image as input to the classification model of Platform 2 for benign and malignant diagnosis. For the second training stage, we fine-tuned the Resnet18 based on the pre-trained model from the first stage, which helped to eliminate the risk of overfitting and improve training accuracy for the classification task. To train Platform 1 and Platform 2, we used Dice loss and Binary Cross-Entropy loss functions, respectively.

B. Performance of the Segmentation Task

In this section, qualitative and quantitative comparisons were performed between the swin-transformer-based lesion segmentation approach in Trans-CEUS and the segmentation algorithms at the forefront of the field. In addition, we conducted the ablation study to verify the effectiveness of the dynamic swin-transformer (DST) encoder, collaborative learning of low-level and high-level features (CLLH) module and spatial temporal separation aggregation (STSA) decoder for lesion segmentation.

1) Comparison With Related Segmentation Methods:

Our proposed Trans-CEUS was compared with several related segmentation methods for dynamic CEUS images including Unet [29], ResUnet [39], Vnet [40], and swin-unet [24]. For different models, we tried the input data with different dimensions, including the 2D, 3D and 2D+T data, respectively in order to explore the best suitable data input mode for the segmentation task of dynamic CEUS. The 2D image frames were stacked along channels for 2D input. The time-dimensioned CEUS image sequence was considered a 3D volume for 3D input. In order to ensure a fair comparison, the total number of frames used for 2D and 3D inputs is identical. For 2D+T input, we extracted the spatial characteristics of the video frame before aggregating the temporal information of those spatial characteristics in the time dimension. Table I displayed the results of CEUS segmentation based on the input dimensions of various models.

TABLE I
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS FOR
THYROID LESION SEGMENTATION

Network	DIM	DICE (%)	IOU (%)
Unet	2D	67.45±7.94	53.85±7.73
ResUnet	2D	70.17±7.62	56.77±7.43
Swinunet	2D	68.29±9.22	55.84±9.10
Unet	3D	71.85±8.16	58.13±8.31
ResU-net	3D	71.54±7.62	57.77±7.43
Swinunet	3D	69.66±8.21	56.04±7.93
Vnet	3D	77.72±7.16	64.59±6.83
Trans-CEUS	3D	78.84±6.71	65.95±7.08
Unet	2D+T	74.98±5.79	60.66±5.63
ResU-net	2D+T	72.59±6.16	58.93±6.51
Swinunet	2D+T	73.72±5.86	59.90±5.49
CEUS-Net	2D+T	78.23±5.44	64.99±5.91
Trans-CEUS	2D+T	82.41±5.13	71.85±5.42

From Table I, we could acquire some interesting observations. 1) Experimental findings comparing input data in 2D, 3D, and 2D+T for the Unet, ResUnet, and Swinunet models revealed that the latter three models performed best when fed 2D+T data for segmentation purposes in real-time CEUS. Thus, 2D+T input which considered the inter-frame temporal information was more suitable for CEUS data. 2) Compared with all segmentation models with 3D input, and classical 3D segmentation model Vnet, our framework achieved better performance of 78.84% using 3D data input. 3) the existing transformer models of swin-unet couldn't be used directly in CEUS lesion segmentation task, because the acquired Dice result of swin-unet was only 73.72% by using 2D+T input. Comparatively, our Trans-CEUS model proposed some novel mechanisms for dynamic CEUS data, and the segmentation result of Trans-CEUS obtained 8.69% improvements in Dice index than that swin-unet. 4) last but not least, compared with the segmentation models with 2D+T input, and the CEUS-net which was specially proposed for CEUS sequence segmentation task, Our Trans-CEUS model achieved the best dice results of 82.41%.

Fig. 6 represented the lesion segmentation results by using different models with 2D+T input qualitatively. In Fig. 6, the segmentation results of 4 benign and 4 malignant lesions were shown, respectively. Our model achieved the highest degree of agreement between modelled and observed lesion regions (red contours). Especially, for the challenging cases that the lesion areas were small (indicated by blue rectangles), the ambiguous boundaries with the surrounding tissue cause the automatic lesion segmentation very difficult [41]. For these challenging cases, although our segmentation result differs from ground truth in some regions, our result is closer to the ground truth in ambiguous boundary regions compared with all other methods.

2) Contributions of DST, CLLH and STSA: Effective feature extraction for input CEUS sequence is achieved through the introduction of the DST encoder and the CLLH module, both of which are described in Section III-B. We used the STSA

decoder approach while on the decoder route of the segmentation challenge. The related ablation experiment was planned as part of this investigation. Firstly, to verify the DST encoder, the encoder from the swin-transformer was replaced by the encoder of Unet3D. Secondly, we also prepared a control experiment in which the CLLH module was intentionally removed. Thirdly, to verify the STSA decoder, the proposed STSA decoder was replaced by the decoder of Unet3D. Please note that, during the ablation study, we also compared the input data with different dimensions, including the 3D and 2D +T data, respectively in order to verify the effectiveness of the 2D+T data input for the segmentation task.

Based on Table II's presentation of matching experimental findings, we could draw the following inferences. 1) without the DST encoder, the segmentation model dropped by an average margin of 6.10% in terms of mean IOU, demonstrating the effectiveness of the DST encoder module. 2) using the CLLH module, our model increased the IOU and Dice results by 1.96% and 0.99%, respectively. 3) the segmentation Dice result could be improved by 1.28%, which validated the effectiveness of the STSA decoder. 4) the 2D+T data input was significant for CEUS segmentation task.

C. Performance of the Diagnosis Task

For demonstration purposes, we compared the Trans-CEUS model with several state-of-the-art deep models in the video recognition field to demonstrate how effective it is for differential diagnosis tasks. Furthermore, we designed the ablation experiments to verify the effectiveness of the variant transformer-based global spatial temporal fusion (GSTF) unit and CNN-based local fine-tuning (LFT) unit which were proposed in the differential diagnosis platform of Trans-CEUS.

1) Network Evaluation of Benign and Malignant Diagnosis: We compared our proposed Trans-CEUS model with the related works. Table III summarized the accuracy, sensitivity, specificity, and FPR by using various algorithms. In addition, the p-values of the prediction results by using the proposed method and other methods were computed to show the significance of the improvement. According to statistical analysis, a p-value smaller than 0.01 indicates a significant improvement. Therefore, the following inferences may be made in light of the experimental outcomes. 1) compared with the direct 3D network (C3D [42] and R3D [43]), the time sequence fusion models (i.e R2plus1D [44], CNN-LSTM [45] and Trans-CEUS) could acquire the significantly better diagnosis accuracy. As an illustration, it was clearly found that the R3D model using 3D CNN performed a mean diagnostic accuracy of 75.26%, whereas the evolution model R2plus1D achieved an accuracy improvement of 2.06%. 2) By a margin of 6.17%, our proposed Trans-CEUS model outperformed the state-of-the-art temporal fusion model CNN-LSTM significantly, achieving the highest average accuracy of 86.59%.

2) Contributions of LFT and GSTF: In this section, we conducted the ablation study to verify the contribution of the CNN based LFT unit and variant transformer-based GSTF unit for the diagnose task. As indicated in Table IV, the diagnostic

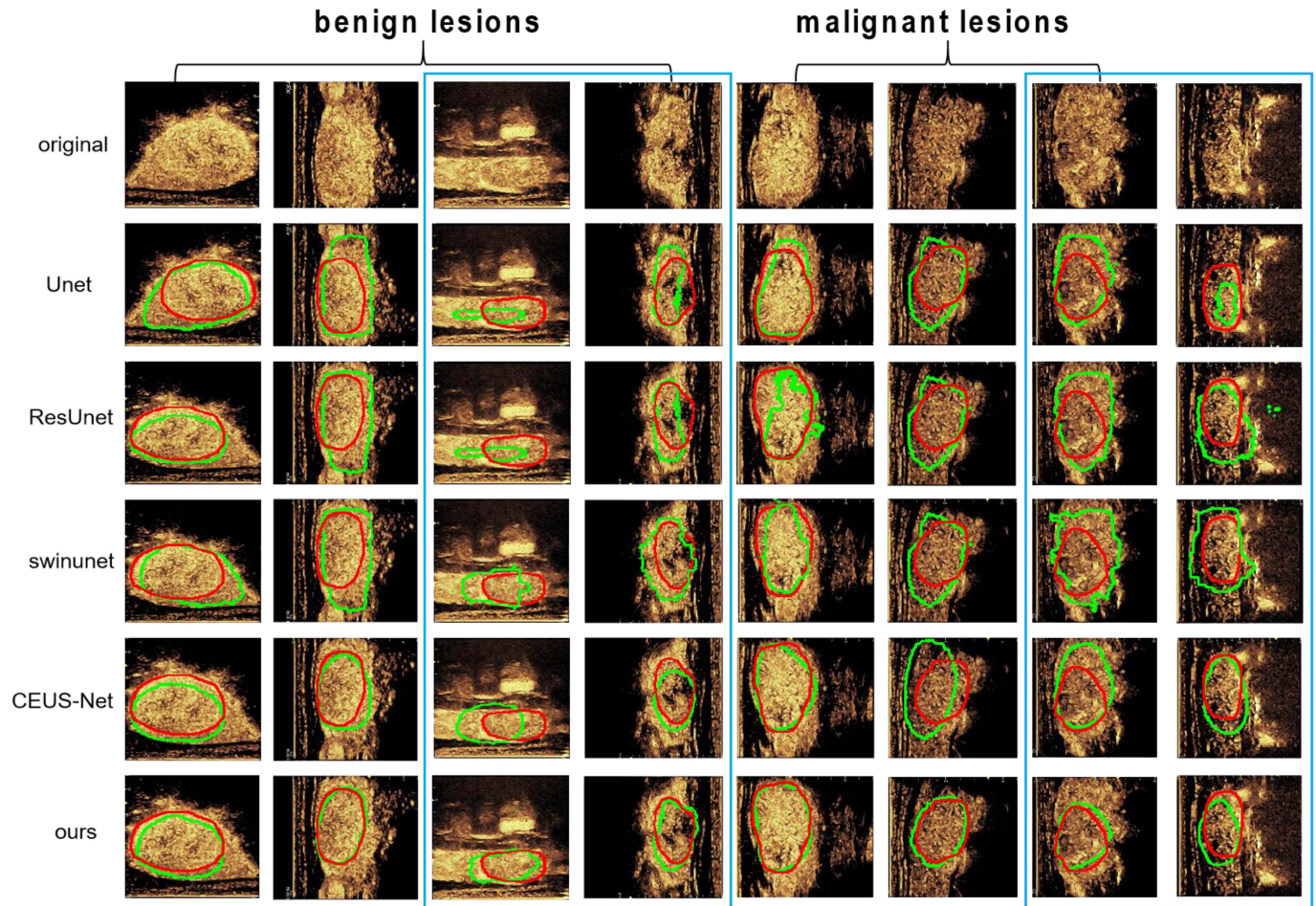


Fig. 6. Segmentation result by using different methods. Green and red contours mean the automatically segmented lesion area and ground truth. Blue rectangles indicate the challenging cases that the lesion areas were small.

TABLE II
QUANTITATIVE EVALUATION RESULTS OF ABLATION EXPERIMENT FOR THYROID LESION SEGMENTATION

Network	DST	CLLH	STSA	DIM	DICE (%)	IOU (%)
Trans-CEUS				3D	74.99 ± 7.81	60.62 ± 8.13
Trans-CEUS	✓			3D	78.01 ± 6.23	64.45 ± 6.70
Trans-CEUS		✓		3D	75.43 ± 5.43	61.97 ± 5.51
Trans-CEUS	✓	✓		3D	78.84 ± 6.71	65.95 ± 7.08
Trans-CEUS				2D+T	76.74 ± 6.99	62.83 ± 7.01
Trans-CEUS	✓			2D+T	80.14 ± 4.76	68.16 ± 5.22
Trans-CEUS		✓		2D+T	77.56 ± 6.68	64.02 ± 6.71
Trans-CEUS	✓	✓		2D+T	81.13 ± 6.01	70.12 ± 6.39
Trans-CEUS	✓	✓	✓	2D+T	82.41 ± 5.13	71.85 ± 5.42

accuracy of the model decreased by 9.79% after the LFT unit was eliminated from the model. This outcome validates the efficacy of our LFT module and suggests that segmentation-specific spatial variables are not transferable to classification problems. On the other hand, the proposed GSTF unit is based on a variant transformer. Therefore, in the ablation study of GSTF unit, we replaced the variant transformer with the convolution operation. Table IV demonstrates that an improvement in average accuracy

of 8.79% was achieved by employing a variation transformer based GSTF unit.

D. Performance of Jointing Segmentation and Differential Diagnosis Tasks

Our Trans-CEUS model proposes the joint learning of the segmentation and differential diagnosis tasks. Specifically, the spatial features extracted for the lesion segmentation task needs

TABLE III
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS FOR THYROID LESION BENIGN AND MALIGNANT DIAGNOSIS

Network	DIM	ACC (%)	Se (%)	Sp (%)	FPR (%)	F1 (%)	p-value (vs. Proposed)
C3D	3D	73.71 ± 4.24	72.16 ± 5.55	75.26 ± 3.09	24.74 ± 3.09	73.30 ± 3.26	p=0.003
R3D	3D	75.26 ± 3.61	76.29 ± 4.12	74.23 ± 3.09	25.77 ± 3.09	75.51 ± 4.12	p<0.0001
R2plus1D	2D+T	77.32 ± 4.64	75.26 ± 4.12	79.38 ± 5.15	20.62 ± 5.15	76.84 ± 4.63	p=0.002
CNN-LSTM	2D+T	80.42 ± 3.93	82.47 ± 6.18	78.35 ± 4.12	21.65 ± 4.12	80.81 ± 3.61	p<0.0005
Trans-CEUS	2D+T	86.59 ± 2.58	87.63 ± 3.09	85.57 ± 2.06	14.43 ± 2.06	86.74 ± 1.54	-

TABLE IV
QUANTITATIVE EVALUATION RESULTS OF ABLATION EXPERIMENT FOR
THYROID LESION BENIGN AND MALIGNANT DIAGNOSIS

LFT	GSTF	Acc (%)	Se (%)	Sp (%)
	√	76.80 ± 5.67	79.38 ± 5.15	74.23 ± 6.18
√		77.80 ± 4.21	77.31 ± 4.12	78.35 ± 5.15
√	√	86.59 ± 2.58	87.63 ± 3.09	85.57 ± 2.06

TABLE V
QUANTITATIVE COMPARISON RESULTS OF DIFFERENTIAL DIAGNOSIS BY
USING THE JOINT TWO-TASK METHODS AND SINGLE DIAGNOSIS TASK
WITH DIFFERENT ATTENTION LAYERS

Method	LAYER	ACC (%)	Se (%)	Sp (%)
Single	1	79.38 ± 5.67	82.47 ± 5.15	76.29 ± 6.18
Single	2	-	-	-
Joint	0	63.40 ± 7.22	59.79 ± 6.19	67.01 ± 8.25
Joint	1	81.96 ± 6.18	85.58 ± 6.18	78.35 ± 6.18
Joint	2	82.47 ± 3.61	81.44 ± 2.06	77.32 ± 5.15
Joint	3	85.05 ± 4.64	86.60 ± 3.09	83.50 ± 6.18
Joint	4	86.59 ± 2.58	87.63 ± 3.09	85.57 ± 2.06

to be fine-tuned in the branch of the classification task. In addition, our model uses the automatically segmented lesion region to promote the diagnosis task. For the purpose of verifying the efficacy of two-task joint learning, we conducted an experiment in which the performance of our joint two-task method was compared to that of single categorization tasks.

Table V not only showed the quantitative results of using two-task joint method and single diagnosis task, but also gave the results of using different number of attention layers in GSTF unit of diagnosis task. From Table V, the following conclusions could be summarized. 1) It was simple to introduce overfitting when employing just the one diagnosis task. Table V's first and two rows reveal that a diagnostic accuracy of 79.38% could be achieved by setting the number of attention layers to 1, whereas setting it to 2 led to an overfitting error. 2) With the joint two-task method, while increasing the number of attention layers from 0 to 4, the diagnosis accuracy also increased. 3) The GSTF unit with four attention layers was able to achieve the best diagnostic accuracy (86.59%) while using the joint two-task technique. Thus, the two-task joint learning could help to finish the differential diagnosis task on the dynamic CEUS dataset.

V. DISCUSSION

Automatic segmentation and diagnosis of benign and malignant by using the dynamic CEUS sequence is a very challenging task. In this section, we first discuss the differences between our proposed Trans-CEUS model compared with the related analysis methods of CEUS, especially some illuminating experimental results. Secondly, we also point out the limitations of our proposed model and propose possible solutions and directions for future research.

A. Model Performance

First, we discuss why our proposed Trans-CEUS model can achieve better performance for dynamic CEUS segmentation task than the existing segmentation methods. As shown in Table I, the existing transformer model swin-unet couldn't acquire good results for CEUS lesion segmentation. Comparatively, our Trans-CEUS model proposed some novel mechanisms for dynamic CEUS data processing. On the one hand, the collaborative learning module of the low-level and high-level features is proposed to eliminate the noise effect of low-level features and help high-level features to better identify the small lesion area. On the other hand, the spatial temporal separation aggregation decoder module which firstly extracts spatial features and then fuses the extracted CEUS spatial feature sequences in the time dimension is proposed to consider inter-frame temporal information for dynamic CEUS analysis. This spatial-temporal modeling and encoder-decoder way may be conducive to lesion segmentation in CEUS.

Second, jointing the segmentation and differential diagnosis tasks has significant impacts on the benign and malignant diagnosis task, as shown in Table V. The segmentation map is used to help the differential diagnosis task to quickly locate the lesions and obtain the region of interest required for benign and malignant diagnosis. The extracted features of the lesion segmentation task can also be fine-tuned for assisting the differential diagnosis. In addition, with the two-task jointing method, the effect of the variant transformer-based GSTF unit is promoted because of the increased diagnosis accuracy. Therefore, the joint learning of segmentation and differential diagnosis tasks can inspire the superiorities of transformer for processing dynamic CEUS data.

From Table I, we could find that 2D+T input was more suitable for CEUS analysis compared with the input data in 2D and 3D. It is thus clear that the inter-frame temporal information is beneficial for dynamic CEUS analysis. This experimental phenomenon is also interpretable from the clinical perspectives.

The CEUS video contains three intervals over time including arterial phase, portal venous and late phases. Some clinical studies have shown that the dynamic perfusion characteristics which are extracted based on the inter-frame temporal information from the 2D+T CEUS sequences may promote the diagnostic performance [46], [47]. For example, the brightness change or inter-frame information of CEUS frames reveals the blood supply of a tumor, and it is important for lesion analysis [18]. Therefore, inter-frame temporal information of 2D+T CEUS input should be considered for dynamic CEUS analysis.

B. Limitations and Future Works

The proposed Trans-CEUS model can finish the automatic lesion segmentation and differential diagnosis of malignant and benign based on CEUS analysis. However, there are still some limitations to be overcome in the future. Firstly, the key image frames of the CEUS video sequence are currently selected by removing redundant frames and sampling with equal time intervals. Automatic extraction of keyframes from US videos of unfixed length begins to attract the increasing attention [48]. Thus, the automatic key frame selection strategy needs to be proposed to simplify the data preparation for CEUS analysis and release a lot of labor consumption. Secondly, due to the need of protecting patient privacy, there is currently no public large-scale CEUS data set. The proposed model is proposed and validated by using the dynamic CEUS dataset only from a single institution. Considering these issues, a federated learning strategy can be used to train the data collected from different hospitals, which can not only expand the data size but also improve the generalization performance of the model. In summary, the developed CEUS model Trans-CEUS not only proposes the swin-transformer encoder mechanism with the collaborative learning of the low-level and high-level features for lesion segmentation from CEUS, but also implements the STSA decoder and variant transformer-based GSTF unit to use the long-distance enhancement perfusion for dynamic CEUS analysis. These improvements can promote the automatic lesion segmentation and differential diagnosis of malignant and benign from dynamic CEUS. The Trans-CEUS model has the potentials to analyze the dynamic CEUS video for various disease such as liver cancer, breast cancer, etc

VI. CONCLUSION

The dynamic CEUS can visualize and quantify tissue perfusion; thus, it is an effective imaging modality to achieve the diagnosis of thyroid nodule. In this paper, we develop a novel CEUS analysis model called Trans-CEUS for finishing the joint learning of automatic lesion segmentation and the differential diagnosis of thyroid nodule. The Trans-CEUS model excavates the long-distance perfusion information of dynamic CEUS, and achieves a good lesion segmentation Dice of 82.41% and superior diagnostic accuracy of 86.59% on dynamic CEUS datasets. Therefore, Trans-CEUS model can acquire accurate segmentation and differential diagnosis results of thyroid nodule. In real clinical applications, the automatic segmentation and diagnosis can help surgeons to analyze lesion and select treatment strategy.

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